Continuous variables + probability as area

1 RECOGNISE IT

- Measurement can take any value on an interval.
- Asked for an interval, tail, density or area.
- A single exact value has probability zero.

2 DECISION / METHOD

1. Name the random variable and its support.
2. Sketch/mark the requested region.
3. Use the distribution CDF or geometric area.
4. Check the answer is between 0 and 1.

3 FORMULA

```
P(a < X < b) = F(b) - F(a)
P(X > a) = 1 - F(a)
P(X = a) = 0
total area = 1
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: uniform and exponential densities, means and variances.
Still know: probability is area; density height is not automatically probability; use complements/differences.

5 EXCEL ROUTE

No universal continuous-probability function. Use the correct family CDF, then:
- left = CDF
- right = 1-CDF
- between = CDF(upper)-CDF(lower)

6 TINY WORKED EXAMPLE

A density is 0.20 on 0 to 5. Then `P(1<X<3)=width x height=2(0.20)=0.40`.

7 TOP EXAM TRAPS

- Adding point probabilities as if X were discrete.
- Treating f(x) as P(X=x).
- Forgetting the support: area outside it is zero.

8 SAY ALOUD

For a continuous variable, probability is area over an interval, and one exact point has probability zero.

9 EXIT CHECK

Can I turn a right-tail or between event into CDF areas without looking?

ONE-GLANCE VISUAL ROUTE

